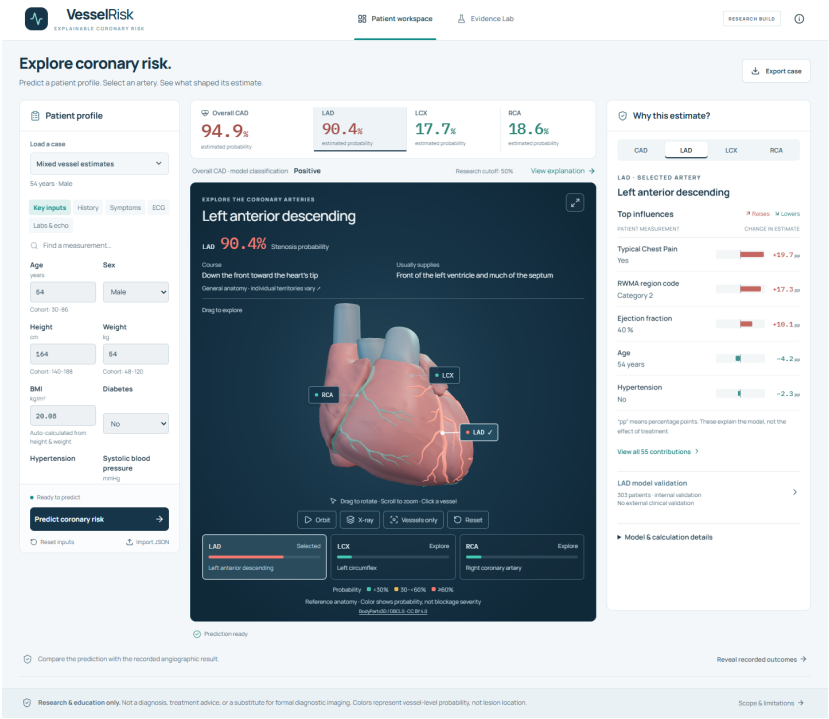
01 / PRODUCT & OBJECTIVE

Coronary risk, made visible.

Four calibrated clinical predictions, independently mapped to real coronary anatomy and explained by patient-level evidence.

VesselRisk connects **structured patient inputs** → **four trained classifiers** → **calibrated probabilities** → **interpretable 3D coronary anatomy**. The first screen pairs the patient profile, heart and target-specific explanation. A separate Evidence Lab exposes validation, calibration, failures, provenance and model hashes.



Actual application screenshot: a held-out cohort case, not a mockup. Outputs are CAD 94.9%, LAD 90.4%, LCX 17.7% and RCA 18.6%. Selected demonstration cases are illustrative, not the evaluation sample.

Track A fit

Official criterion	Delivered evidence
Prediction · 30%	Four targets; nested repeated CV; model/calibration comparisons; honest per-target metrics
3D · 25%	63 licensed meshes; separate coronary territories; rotate, zoom, select, X-ray and vessel isolation
Interpretability · 20%	Exact-probability SHAP reconstruction, measured inputs, global ranks, model disagreement
Integration · 15% / technical · 10%	Validated API; synchronized outputs; local assets; pinned artifacts, hashes, tests and startup

02 / DATA & SCIENTIFIC PROTOCOL

Validation without shortcuts.

Target columns are rejected during fitting, preprocessing and inference.



Selection, preprocessing and calibration use outer training data only; test labels stay isolated.

Source workbook	Actual inspection
Populated worksheet	303 patients x 59 columns; 55 inputs. Empty Sheet1 excluded.
Quality audit	0 missing / duplicate rows; constant Exertional CP retained.
Positive / negative	CAD 216 / 87; LAD 177 / 126; LCX 119 / 184; RCA 114 / 189
Label discrepancy	Patient row 94: Cath Normal, LAD Stenotic. Source labels preserved.

Repeated nested stratified cross-validation

Outer: stratified 5-fold x 2 repeats, seed 2026; average two held-out probabilities per patient. **Inner:** three folds, six fixed candidates; maximize ROC-AUC - 0.25 x Brier. Select parameters, then family.

Family	Search / fixed configuration
Logistic regression	C = 0.1 or 1; balanced class weights; max_iter 2500
Random forest	min_samples_leaf = 3 or 8; 120 trees; max_features .7; balanced_subsample
XGBoost	max_depth = 2 or 3; 110 trees; rate .04; lambda 5; min_child_weight 4

Fold-local pipeline: numeric median imputation/scaling; categorical mode imputation/one-hot. No global selection, SMOTE or merging. Class weights are training-only; XGBoost uses regularization.

Calibration: predeclared sigmoid; three-fold OOF within outer train, then base-estimator refit (`ensemble=False`). Raw/isotonic: sensitivity only. Full-cohort artifacts use inner selection and refit.

Scope: headline metrics average two held-out models per patient; the workspace explains one fitted model. No external validation. Operating points appear separately (page 3).

03 / MEASURED PREDICTIVE PERFORMANCE

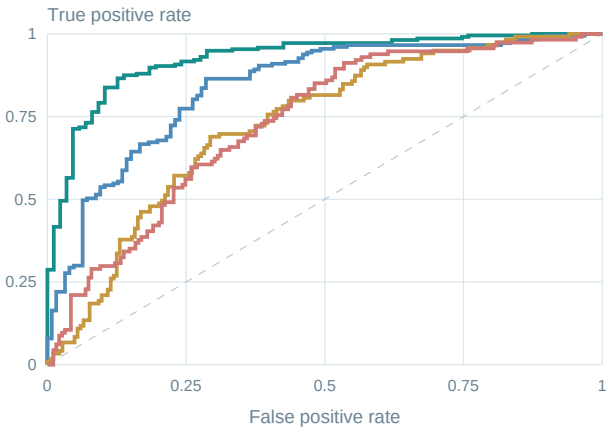
Strong CAD; harder vessel tasks.

Primary metrics use one repeat-averaged held-out probability per patient. Scores below are from shipped artifacts.

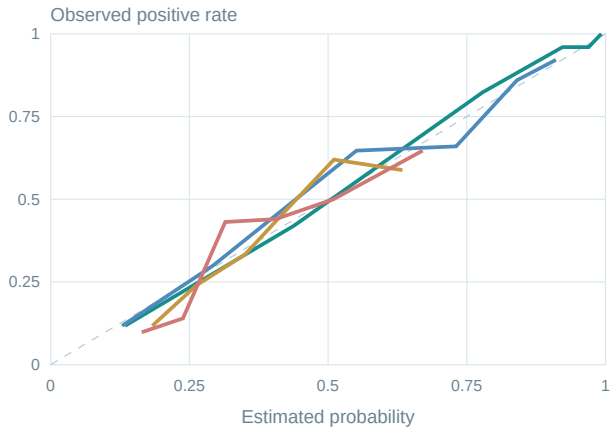
Target	AUC [95% CI]	Acc	Prec	Recall	F1	AP	Brier ↓
CAD	0.925 [0.891, 0.954]	0.868	0.896	0.921	0.909	0.967	0.094
LAD	0.839 [0.791, 0.886]	0.789	0.793	0.864	0.827	0.868	0.157
LCX	0.728 [0.670, 0.786]	0.673	0.625	0.420	0.503	0.575	0.206
RCA	0.732 [0.672, 0.790]	0.660	0.570	0.395	0.466	0.577	0.201

Intervals: 1,000 patient-stratified bootstrap resamples conditional on fitted folds. They omit retraining/selection uncertainty and are not external-validation intervals. Fold SD is descriptive because folds overlap. AP is average precision.

ROC / all four targets



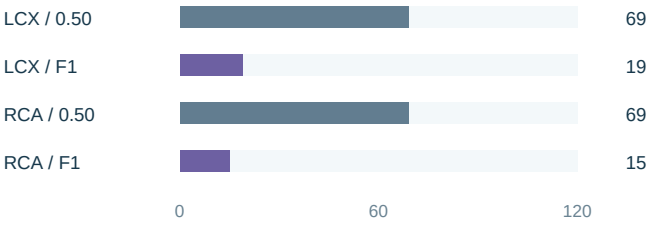
Calibration / six quantile bins



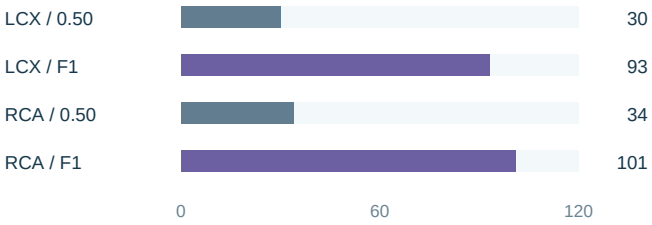
CAD LAD LCX RCA

Training-selected research operating points

Missed positives (patients)



False positives (patients)



Gray: fixed 0.50; purple: training-selected F1. Same models/probabilities. LCX recall .420 to .840 (119 positives); RCA .395 to .868 (114 positives). Negative counts: 184 / 189. F1 cutoffs use inner OOF from outer train only; pool mean probability-minus-cutoff per patient. AUC/Brier stay unchanged. Accuracy falls .673 to .630 / .660 to .617. Research tradeoffs, not clinical cutoffs.

Neither the earlier ExtraTrees/SVM expansion nor the final elastic-net/HistGradientBoosting/CatBoost search showed a material policy gain. Final LCX/RCA AUC: .731/.733 versus .728/.732; paired delta intervals include zero. Original models retained. Reused splits are development evidence, not independent confirmation; full trial logs: experiments/final_audit.

04 / INTERPRETABILITY & UNCERTAINTY

Explain the actual probability.

A contribution must explain the value the user sees, not an unrelated raw model margin.



Base + all signed contributions reconstruct the displayed calibrated probability.

Local SHAP: permutation explanations use the selected calibrated classifier and training-only background. Original categorical variables remain indivisible coalitions. Two cycles, 16/20 background patients, all 55 contributions. Reconstruction to 1e-6 checks additivity, not attribution precision.

Held-out global explanation evidence

Target	Leading absolute probability contributors	Rank stability
CAD	Typical Chest Pain, Age, Hypertension, Diabetes	0.761
LAD	Typical Chest Pain, RWMA region code, Ejection fraction, Age	0.775
LCX	Age, Typical Chest Pain, Creatinine, Platelets	0.750
RCA	Age, Typical Chest Pain, ESR, Diabetes	0.730

Global importance: mean absolute probability SHAP, six held-out patients x ten folds; background n=16, one cycle. Stability: mean pairwise Spearman correlation of fold rankings. Approximate, descriptive and noncausal; correlated features can share credit.

Three complementary limits on confidence

Signal	Meaning and limitation
Candidate disagreement	Probability spread across three families exposes sensitivity to model choice. Not a confidence interval.
Refitted model range	Min/max of ten outer models; demo uses two excluding that patient. Descriptive; may omit the full-cohort estimate.
Profile representation	Standardized mixed-feature nearest-neighbor distance, cohort leave-one-out 95th percentile + single-feature range checks. Unvalidated heuristic.

Inspect failures, not only successes

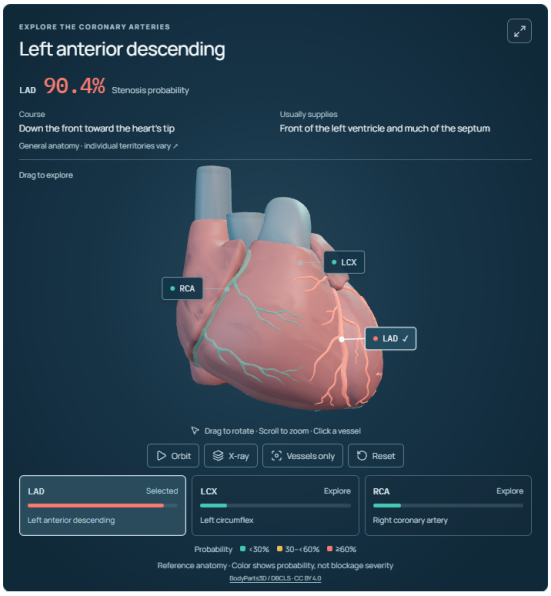
Evidence Lab exposes sex/age subgroups, eight largest held-out errors per target and fixed-logistic 5-fold ECG/laboratory-echo ablations. Small samples support descriptive checks, not fairness certification. The demo includes a confident CAD miss; the source discrepancy remains visible.

Boundary: editing inputs explores model response, not treatment effect. Independent CAD/vessel classifiers may disagree; the UI exposes this. Attribution does not establish causality or clinical validity.

05 / 3D & END-TO-END INTEGRATION

Real anatomy, honest mapping.

A memorable interaction with traceable structure identity, not invented lesion localization.



Actual interactive viewer. Heart-wall meshes, great vessels and independently selectable coronary branches are loaded locally; no asset network dependency during the demo.

Vessel mapping	Source geometry / identity
LAD	Anterior interventricular / anterior descending branches; trunk FJ2631
LCX	Circumflex branches of left coronary artery; FJ2649-FJ2654
RCA	Right coronary artery trunk and branches; trunk FJ2723

Asset: BodyParts3D 4.0, DBCLS, CC BY 4.0; pinned human-atlas derivative. 63 meshes, 37,444 triangles, 1,071,048 bytes. Selected, recentered and uniformly scaled; source names/IDs retained. Generic adult male anatomy, not a patient image. No RWMA-to-lesion mapping or geometry deformation.

Interaction: rotate, zoom, pan, select vessel, orbit, X-ray, vessel isolation, reset and expanded view. Artery colors use model probabilities; projected labels connect to artery surface vertices. The artery guide explains typical course and supply, not patient injury. The renderer limits pixel ratio to 1.5. Software WebGL browser tests exercise the interface without a dedicated GPU; universal FPS is not claimed.

System path: Next.js clinical form → validated FastAPI request → calibrated CAD/LAD/LCX/RCA models → SHAP/uncertainty/OOD → probability, anatomy and explanation updates. Edited inputs mark previous results and explanations stale and block stale exports. API failures preserve the previous result and explain how to retry. Height and weight remain editable; BMI is derived. Cohort defaults are explicitly marked.

06 / REPRODUCIBILITY, USAGE & LIMITS

Ready to run and review.

The software, model weights and evidence run locally without cloud credentials or inference APIs.

Run the complete system

Python 3.13 and Node 24 recommended. First run: `python scripts/start.py --setup`. Subsequent launches: `python scripts/start.py`. Open **http://127.0.0.1:3000**. Select a held-out case → Predict → select vessels → inspect explanations → Evidence Lab. Optional Docker Compose packaging is supplied; Docker execution was not verified on this host.

Reproduce / inspect	Command or artifact
Train and pair vessel audit	python -m ml.train (seed 2026; 5 x 2 outer folds) python -m ml.refine_vessels
Verify frozen OOF results	python -m ml.evaluate → experiments/metrics.csv
Critical validity checks	python -m pytest
Frontend checks	cd web; npm run typecheck; npm run lint; npm run build
Browser demo reliability	cd web; npx playwright install chromium; npx playwright test
Scientific artifacts	ml/artifacts/{CAD,LAD,LCX,RCA}.joblib; evidence.json; registry.json
Submission materials	README; data/model card; licenses; screenshots; five-minute demo script

Scope and limitations

Small single-center referral cohort with high CAD prevalence; no external clinical validation or longitudinal outcome prediction. Independent classifiers do not produce a validated joint probability. Vessel estimates are weaker, with low recall at the default threshold. Mixed-feature OOD and refit ranges are descriptive heuristics. No diagnosis, treatment advice, stenosis severity or lesion coordinates. Human review is required before submission; the final YouTube recording/upload uses the submitter's account.

Sources, licenses and AI disclosure

Official task: supplied **Track A.pdf**. Clinical data: Alizadehsani, Roshanzamir & Sani, UCI Extension of Z-Alizadeh Sani, **DOI 10.24432/C5461K**, CC BY 4.0. Anatomy: **BodyParts3D / DBCLS**, CC BY 4.0; Mitsuhashi et al., DOI 10.1093/nar/gkn613; derivative pinned to human-atlas commit 1c38bf35c254a891200d3cedecfd57abebe83d8d. Original application code MIT; Manrope and IBM Plex Mono fonts OFL. Detailed attribution and dependency terms: ATTRIBUTIONS.md.

Method references: scikit-learn nested CV and probability calibration documentation; SHAP permutation explanations. Exact URLs and prior-winner research are in docs/RESEARCH.md. The six-page limit applies to this report; full audit tables remain in the software and artifacts.

AI assistance: OpenAI Codex assisted research, coding, debugging, verification and drafting. The deployed product uses no generative model to predict, diagnose or advise. All reported model measurements are computed from the supplied cohort, not generated or copied.